

MERN Stack Intern & Junior Developer Interview Questions

Section 1: Introduction (5%)

1. Introduce yourself.
2. Tell us about your MERN project.
3. What was your role in the project?
4. What challenges did you face and how did you solve them?
5. Which part of MERN do you enjoy the most?

HTML & CSS (5%)

1. Difference between `id` and `class`.
2. What is Flexbox? When do you use it?
3. Difference between Grid and Flexbox.
4. What is responsive design?
5. Explain the CSS Box Model.
6. What is semantic HTML?
7. How do you center an element?

Practical

- Build a responsive navbar.
- Create a login page.

JavaScript (20%)

Fundamentals

1. Difference between `var`, `let`, and `const`.
2. Explain scope.

3. Explain closure.
4. What are callbacks?
5. What is a Promise?
6. async/await vs Promise.
7. Difference between `==` and `===`.
8. Explain hoisting.
9. Explain event loop.
10. Difference between synchronous and asynchronous programming.

Arrays

1. `map()`
2. `filter()`
3. `reduce()`
4. `find()`
5. `some()`
6. `every()`

Objects

1. Destructuring
2. Spread operator
3. Rest operator

ES6

1. Arrow functions
2. Template literals
3. Modules
4. Default parameters

Practical

- Reverse a string.
- Remove duplicate array values.

- Find the second largest number.
- Group users by age.
- Flatten nested arrays.
- Debounce a search function.

React.js (25%)

Basics

1. What is React?
2. Virtual DOM.
3. JSX.
4. Components.
5. Props.
6. State.
7. Functional vs Class components.
8. Controlled vs uncontrolled components.

Hooks

1. useState
2. useEffect
3. useRef
4. useMemo
5. useCallback
6. Custom Hooks

Routing

1. React Router
2. Dynamic Routes
3. Nested Routes
4. Protected Routes

State Management

1. Context API
2. Redux Toolkit
3. When should Redux be used?

Performance

1. React.memo
2. Lazy Loading
3. Code Splitting

Forms

1. Form validation
2. React Hook Form
3. Yup validation

Practical

- Build CRUD using React.
- Create a search filter.
- Pagination.
- Infinite scrolling.
- Fetch API data.
- Login page with validation.
- Dark mode.
- Shopping cart.

Node.js (15%)

1. What is Node.js?
2. Event loop.

3. Streams.
4. Buffers.
5. File System module.
6. Path module.
7. Express.js.
8. Middleware.
9. Authentication middleware.
10. Error handling.

Practical

- Build REST API.
- Upload files.
- Send email.
- Logging middleware.

MongoDB (15%)

1. SQL vs MongoDB.
2. Collections vs Documents.
3. ObjectId.
4. Aggregation.
5. Indexing.
6. Embedding vs Referencing.
7. Schema validation.

Practical

- CRUD operations.
- Pagination.
- Search users.
- Filter by multiple fields.

- Aggregate total orders.

API (10%)

1. REST API principles.
2. HTTP methods.
3. Status codes.
4. JWT Authentication.
5. Refresh Token.
6. CORS.
7. Postman usage.

Practical

- Login API.
- Register API.
- Protected API.
- Upload image API.

Git & Deployment (5%)

1. Git workflow.
2. Merge vs Rebase.
3. Resolve merge conflicts.
4. GitHub Pull Request.
5. Environment variables.
6. Production deployment.

Database Design

1. Design an e-commerce database.
2. Design a blog database.
3. Design a school management database.
4. How would you store orders?
5. How do you prevent duplicate data?

Debugging Questions

1. API returns 500. What will you check first?
2. React page is blank. How do you debug it?
3. JWT authentication fails. What could be wrong?
4. MongoDB connection fails. How do you investigate?
5. API works locally but not on the server. What steps do you take?

Project-Based Questions

1. Explain your project architecture.
2. Why did you choose MongoDB?
3. How is authentication implemented?
4. How do you protect routes?
5. How do you handle validation?
6. How do you upload images?
7. How do you manage state?
8. What improvements would you make with more time?
9. What feature are you most proud of?
10. What would you change if you rebuilt the project?

Practical Coding Round (Highly Recommended)

1. Build a Login API with JWT.
2. Create a React Login Form with validation.
3. Build a CRUD application.
4. Create a Todo application with filtering.
5. Implement pagination in React.
6. Create a search API with MongoDB.
7. Upload an image using Multer.
8. Build a protected dashboard.
9. Integrate frontend with backend APIs.
10. Deploy the application and explain the deployment process.